

Bidirectional EN ↔ ES translation transformer

ENCODER

DECODER

Reading the diagram

— data flow

- - - residual connection

- - - encoder to decoder

37.6M parameters

4 encoder + 4 decoder layers

d_model 512, 8 heads, d_ff 2048

one shared 16k vocabulary

embeddings tied 3 ways

the only path from source to target

memory
(6, 512)

K, V

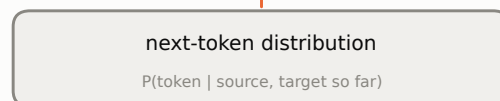
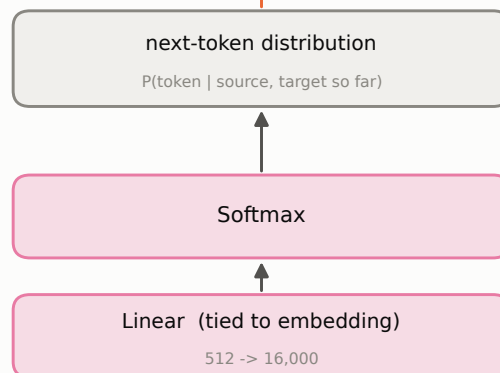
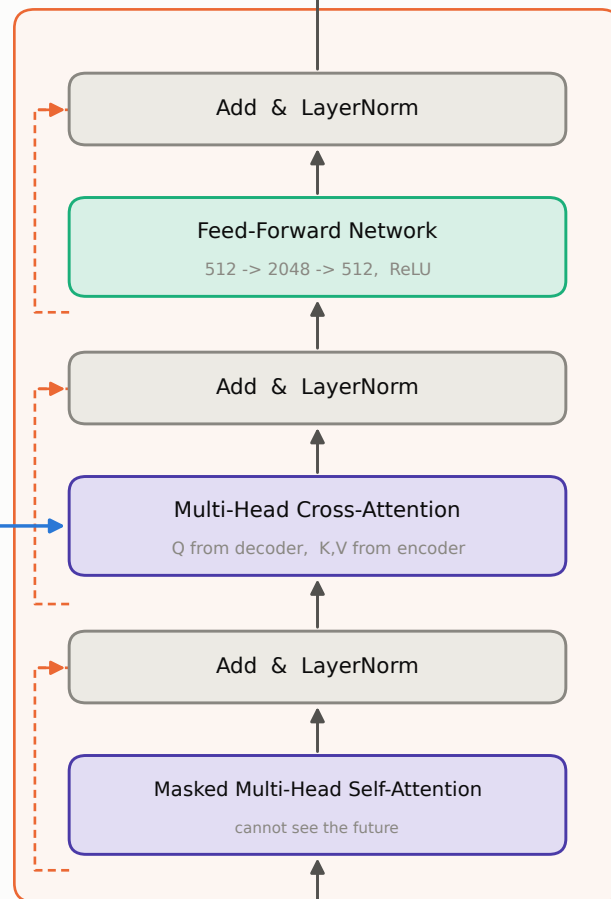
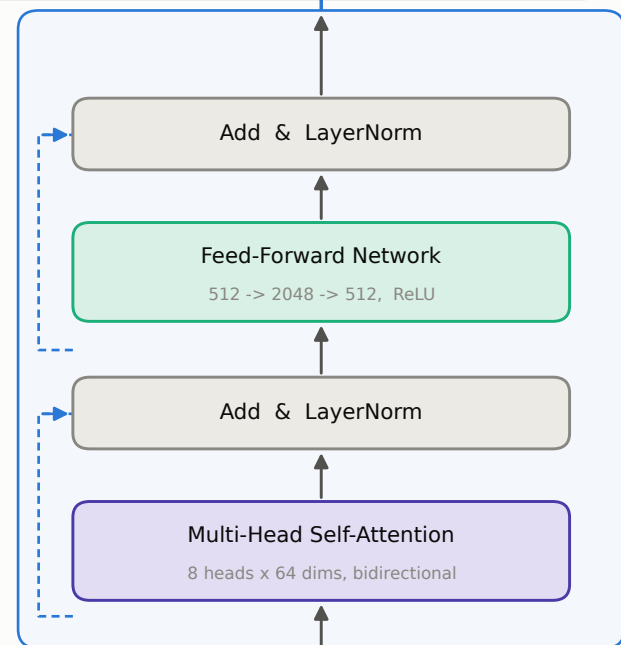
4x

source: <2es> I am tired . </s>

the direction tag tells one model which way to translate

target so far: <s> Estoy cansado .

shifted right: position t predicts token t+1



Estoy cansado . </s>